

KHALLAF TAHER

Cloud & IoT Engineering Student · HENNGE GIP Candidate

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EDUCATION

Institut National des Postes et Télécommunications (INPT)

Rabat, Morocco | 2025 – Present

Engineering Degree in Cloud & IoT Engineering

The University of Tokyo (UTokyo GCI)

Tokyo, Japan | 2026 (Enrolled)

Global Consumer Intelligence & Data Science Program

Classes Préparatoires Ibn Timiya

Marrakech, Morocco | 2023 – 2025

Intensive Prep Curriculum in Mathematics, Physics & Computer Science

TECHNICAL SKILLS & LANGUAGES

Cloud & DevOps: AWS (EC2, S3, IAM), Docker, Docker Compose, Terraform (IaC), GitHub Actions (CI/CD), Nginx, Linux (Bash)

Backend & Web: Python (FastAPI, PyTorch), React, JavaScript (ES6+), REST APIs, OAuth2, JWT (Argon2/Bcrypt), HTML/CSS

DB & Protocols: PostgreSQL, SQLAlchemy ORM, Alembic, ONNX Runtime, Mosquitto MQTT, slowapi Rate Limiting, Git

Languages: Arabic (Native), English (Fluent / Debater), French (Fluent), Japanese (Basic / Self-Studying)

TECHNICAL PROJECTS

Cloud Notes API & AI RAG Assistant [\[Live API Docs\]](#)

Personal Project | 2025

Tech: FastAPI, PostgreSQL 15, Anthropic Claude 3.5 Sonnet, Docker, AWS EC2, Terraform IaC, GitHub Actions

- Engineered a containerized backend on AWS EC2 behind an Nginx SSL proxy with stateless JWT auth (Argon2).
- Integrated Claude 3.5 Sonnet RAG vector search (/ask_notes) for natural language querying & automatic note summarization.
- Enforced system security using slowapi rate limiting (10 req/min) and automated AWS deployment via Terraform & CI/CD.

ECG Multimodal Edge Inference Chain

Supervised Internship Project | 2025

Tech: PyTorch, ONNX Runtime, INT8 Static Quantization, Raspberry Pi 4, Mosquitto MQTT, WFDB Chapman Dataset

- Optimized a 12-lead ECG neural model (1D-CNN + BiLSTM) trained on 10,500+ records for embedded Raspberry Pi 4 execution.
- Quantized PyTorch weights to ONNX INT8 static calibration, cutting latency to **86.31 ms** (8% faster) with **98.93% Accuracy**.
- Architected a real-time MQTT telemetry pipeline publishing live edge predictions to Mosquitto broker clients.

3D IoT Digital Twin Engine & AWS Telemetry [\[Live 3D App | API Docs\]](#)

Personal/Coursework Project | 2026

Tech: Three.js (3D WebGL), WebSockets, AWS IoT Core TLS, AWS RDS PostgreSQL 16, FastAPI, Redis, Docker

- Architected a real-time 3D WebGL Digital Twin platform with Three.js and WebSockets for dynamic sensor state visualization.
- Configured AWS IoT Core MQTT bridge over TLS (X.509 certs), live cloud database on AWS RDS PostgreSQL 16, and Redis state caching.
- Implemented bi-directional actuator remote control APIs and an automated threshold evaluation alert engine.

Post Management REST API

Personal Project | 2025

Tech: Python, FastAPI, PostgreSQL, SQLAlchemy ORM, Alembic, JWT, OAuth2

- Developed a secure RESTful API with full CRUD functionality, voting system, and OAuth2/JWT authorization scoping.
- Managed database schemas and seamless automated migrations using SQLAlchemy ORM and Alembic.

LEADERSHIP & CERTIFICATIONS

- Head of English Cell — Great Debaters Club (INPT):** Designed and led a 14-session onboarding program teaching British Parliamentary debating, evidence-based argumentation, and public speaking.
- Organizer — Forum INPT 2025:** Coordinated student-company relations for Morocco's 21st national engineering forum.
- Certifications:** Oracle OCI AI Foundations Associate (2025) · Oracle OCI & AWS Architect Formation (2025) · Stevens Initiative Global Leadership (ASU & Lycée d'Excellence, 2022).